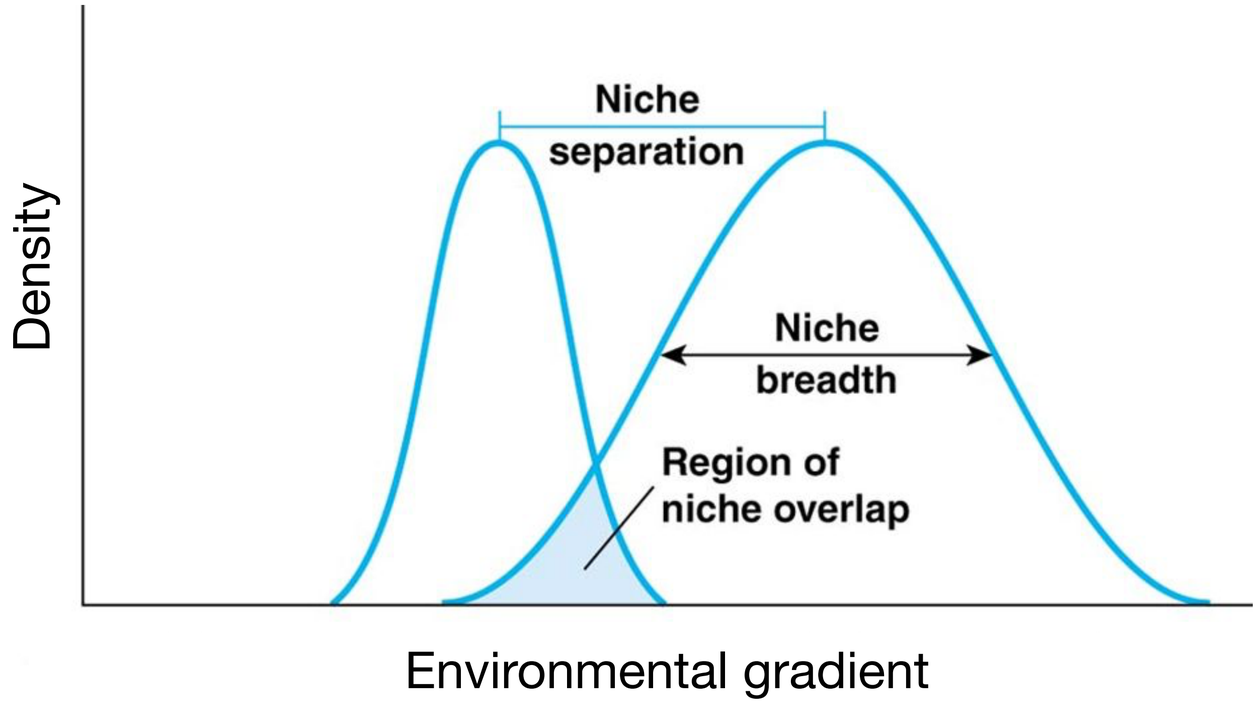
Real-World Applications

Modeling the distribution of biodiversity under global climate change

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Niche overlap estimates



Niche overlap

Niche overlap: The degree to which two species share the same environmental conditions or resources—i.e., how closely their ecological requirements coincide. It allows researchers to quantitatively compare ecological niches.

Allow researchers to quantify and compare environmental niches.

Why Study Niche Overlap?

Species coexistence and competition: High overlap between species can indicate strong potential for interspecific competition.

Evolutionary processes (e.g., sister species): Did lineages diverge due to distinct environmental pressures (ecological speciation), or did they retain ancestral niches (niche conservatism) and speciate via other mechanisms (e.g., allopatric/ reproductive isolation)?

Biological invasions: Comparing a species' niche between its native and invaded ranges reveals whether it has undergone a niche shift or if it has yet to fully colonize available habitat in the novel range.

Approaches to measuring niche overlap

Geographic overlap: Comparing spatial distribution maps (geographic space).

Physiological responses: Mechanistic testing via laboratory or field experiments.

Environmental overlap: Quantifying shared occupancy in multivariate environmental space.

Environmental overlap

Quantitative metrics that measure the degree to which the niches of two species overlap based on their environmental distributions.

Schoener's D: Measures shared environmental space based on direct probability differences, giving greater weight to high-suitability areas where species are most likely to occur.

Hellinger's I: Applies a square-root transformation to probability distributions, reducing the influence of extreme suitability values for a more balanced comparison.

Both D and I range from 0 (no overlap) to 1 (complete overlap), allowing straightforward interpretation and comparison.

Warren et al., 2008

Environmental overlap

Niche Equivalency Test: Are the two niches ecologically identical (interchangeable)?

Similarity test: Are the two niches more similar than expected by chance, given the available background environment?

Interpretation ($p < 0.05$): The observed difference (equivalency test) or similarity (similarity test) is statistically significant and unlikely to have occurred by random chance.

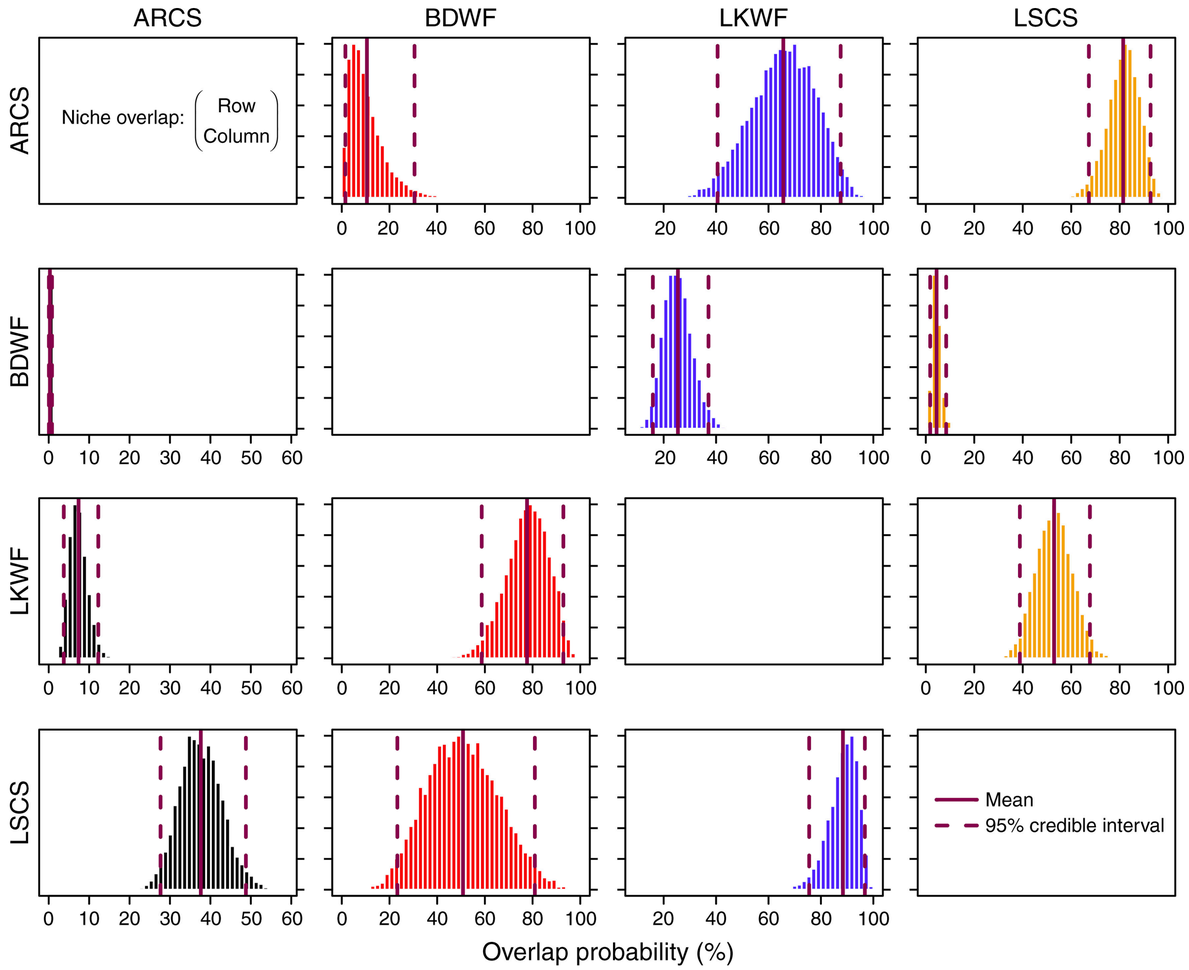
Probabilistic method for quantifying n-dimensional niches

Defines overlap as the probability that an individual from Species A occurs within the niche region of Species B.

Directional (asymmetric), multivariate, and grounded in a Bayesian framework.

Directly aligned with Hutchinson's n-dimensional hypervolume concept.

Swanson et al., 2015



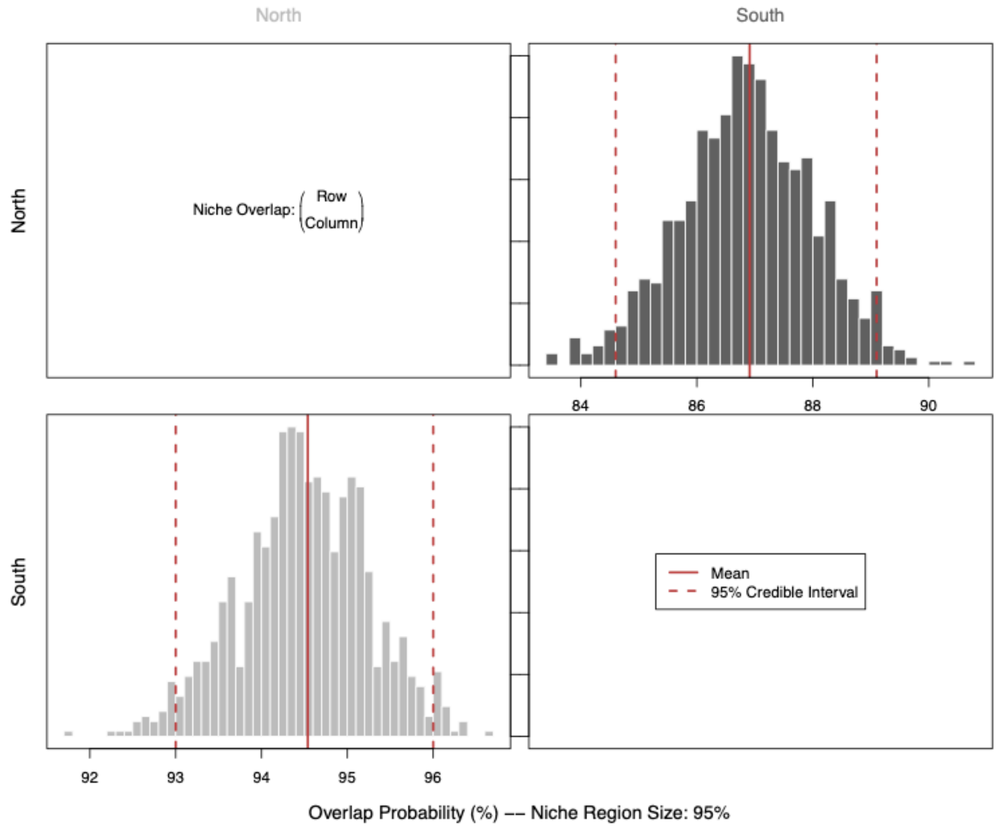
Distribution of the probabilistic niche overlap metric (%) for a specified niche region - **probability of species displayed in rows overlapping onto those displayed in columns**. Mean and 95% intervals of niche overlap in purple.



Global distribution

Niche overlap between the two major genetic groups of *Macrocystis pyrifera*

Assis et al., 2023



The posterior distribution of the probabilistic niche overlap between the northern and southern hemispheres was 86.88%, while the overlap between the southern and northern hemispheres was 94.59%.

	D	I
Observed	0.550	0.705
Equivalency test (p-value)	1	1

Equivalency Test: Failed to reject the null hypothesis ($p > 0.05$) - the northern and southern groups occupy statistically indistinguishable ecological niches.

Conclusion: Together, these results provide strong evidence for niche conservatism across hemispheres in *Macrocystis pyrifera*.